
Machine Learning Basics

An introduction to ML concepts, algorithms, and model training with scikit-learn.

Level: Advanced | Estimated time: 15 hours | Category: Data

Course overview

An introduction to ML concepts, algorithms, and model training with scikit-learn.

This PDF course pack was generated for the project from publicly available internet resources and the app's built-in lesson outline.

Lesson 1: Introduction to Machine Learning

Machine learning enables computers to learn from data without explicit programming.

- Supervised vs. unsupervised vs. reinforcement learning
- Classification vs. regression problems
- Training set, validation set, test set
- Overfitting and underfitting
- Common algorithms: Linear Regression, Logistic Regression, Decision Trees, KNN, SVM
- Feature engineering and scaling
- Model evaluation: accuracy, precision, recall, F1, ROC-AUC
- scikit-learn pipeline: fit, transform, predict

Key concept: The goal of machine learning is to find patterns in data that generalize well to new, unseen data.

Further reading from the internet

These official or widely used references match the course topic and can be used for deeper study.

scikit-learn getting started

https://scikit-learn.org/stable/getting_started.html

scikit-learn supervised learning

https://scikit-learn.org/stable/supervised_learning.html

scikit-learn clustering overview

<https://scikit-learn.org/stable/modules/clustering.html>